



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA-0211 C-PROGRAMMING

Distributed Border Surveillance System with Duplicate Detection and Secure Alert Processing

Guided by:

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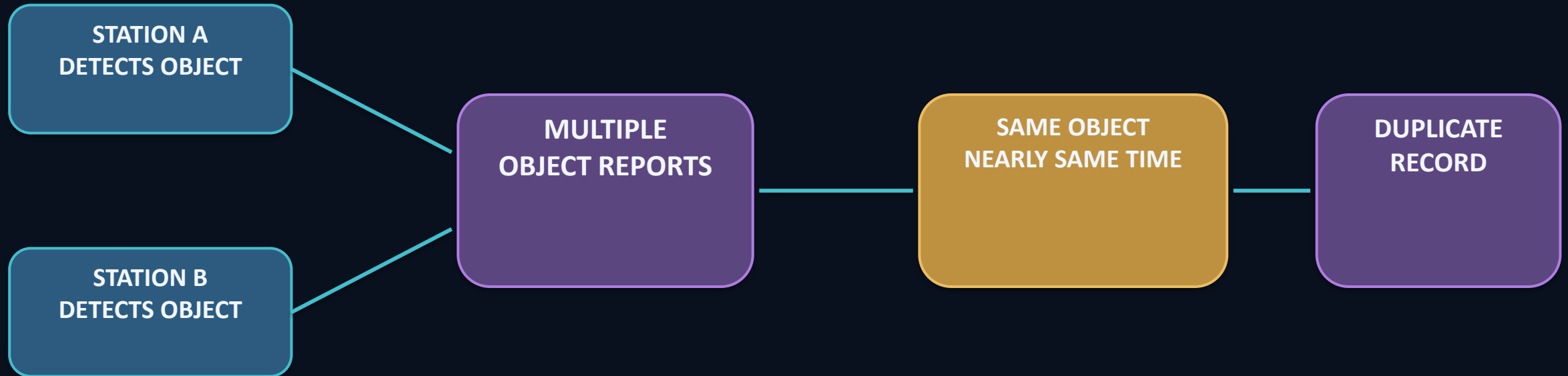
Presented by :

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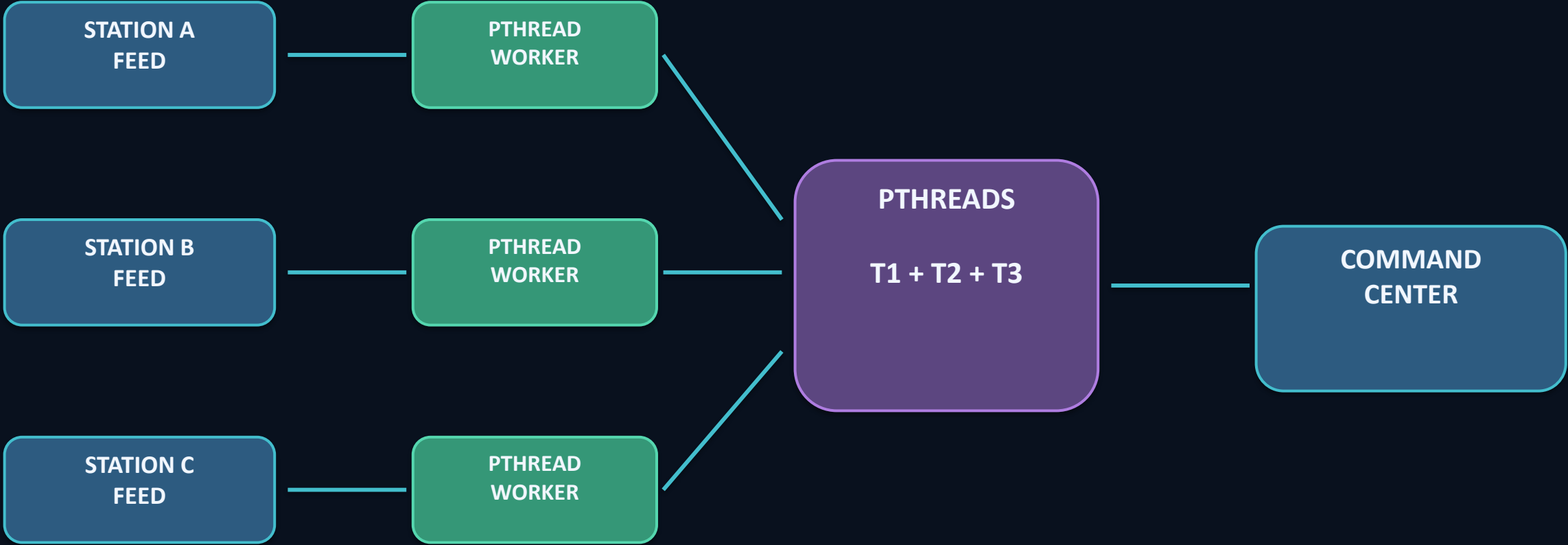
STEP 1 — IDENTIFY THE PROBLEM



PROBLEM → THE SYSTEM MAY COUNT ONE REAL OBJECT MORE THAN ONCE



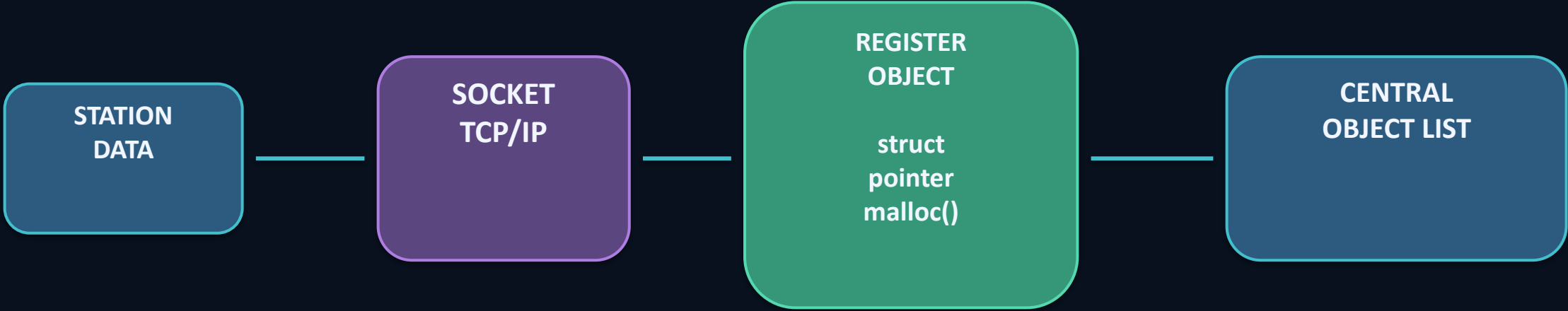
STEP 2 — COLLECT FEEDS IN PARALLEL



METHOD → PTHREADS HANDLE MANY STATION FEEDS WITHOUT WAITING FOR ONE FEED TO FINISH



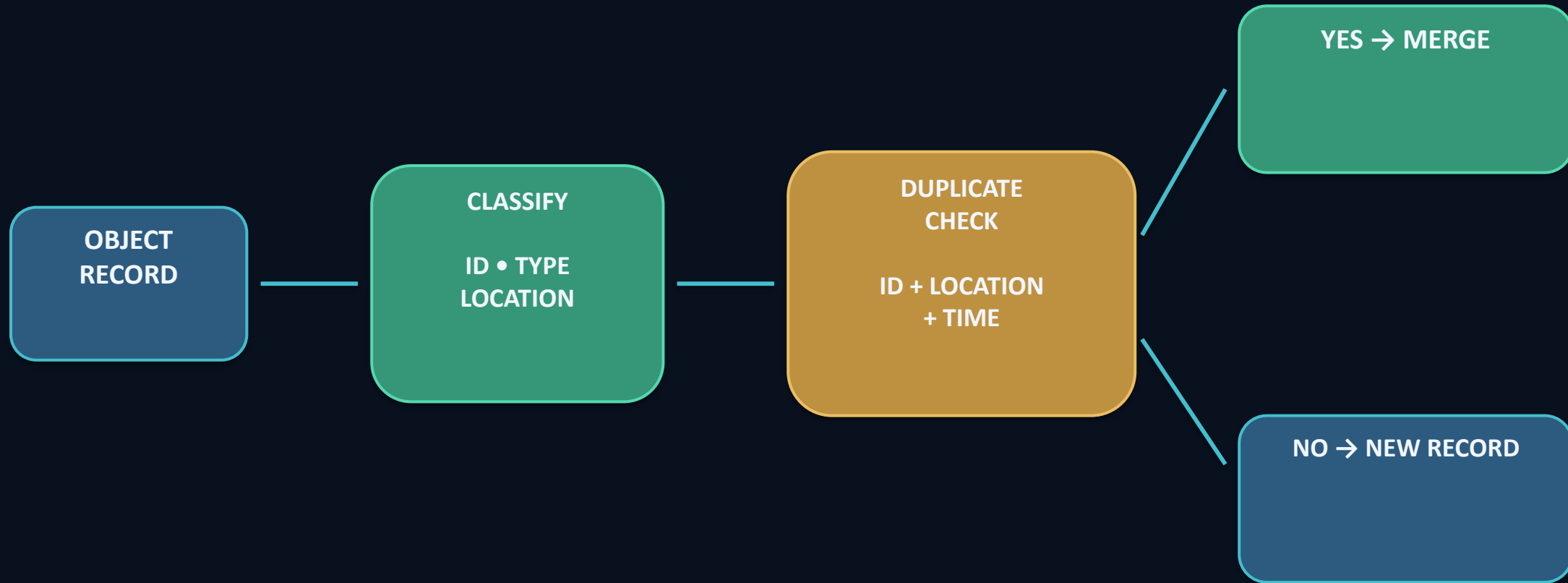
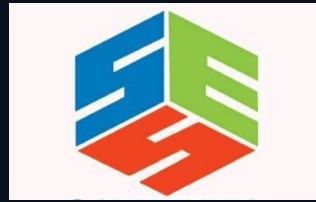
STEP 3 — SEND & REGISTER OBJECT DATA



METHOD → SOCKETS TRANSFER DATA; DYNAMIC MEMORY CREATES RECORDS AS OBJECTS ARRIVE



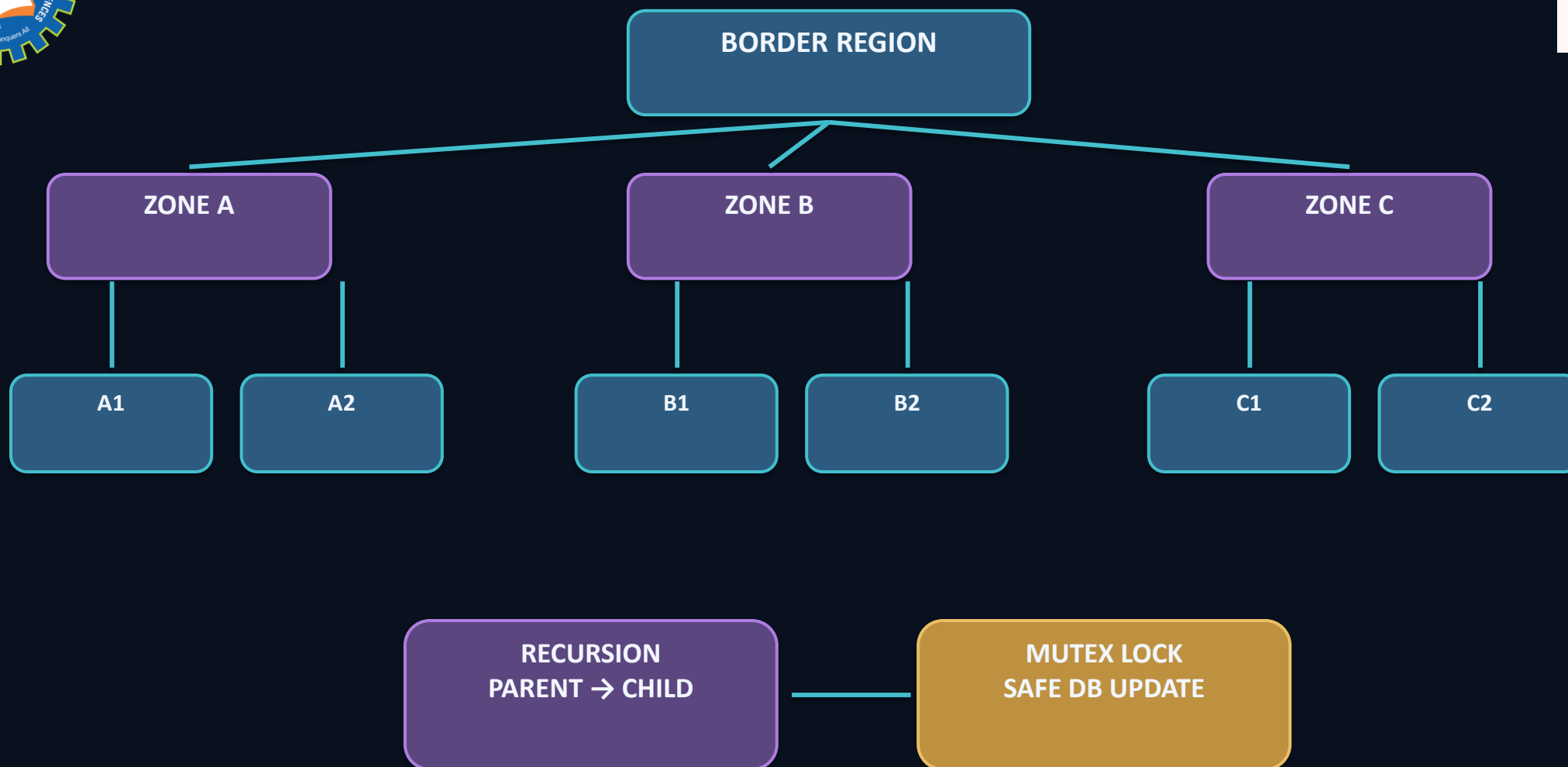
STEP 4 — CLASSIFY & REMOVE DUPLICATES



KEY SOLUTION → TWO NEAR-SIMULTANEOUS REPORTS ARE MATCHED AND MERGED INTO ONE RECORD



STEP 5 — SEARCH ZONES & SAFELY UPDATE DATABASE



METHOD → RECURSION VISITS HIERARCHICAL ZONES; MUTEX PREVENTS CONCURRENT DATABASE CONFLICTS



STEP 6 — LOG, ALERT & FINAL SOLUTION



FINAL CONNECTION → DETECT → RECEIVE → REGISTER → DEDUPLICATE → SYNCHRONIZE → LOG/ALERT → RESPOND